The Impact of the Cost-Of-Living Crisis on Travel Choices of Scottish Residents: An Exploratory Analysis

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ABSTRACT

This study aims to identify the impact of the cost-of-living crisis on travel choices of Scottish residents. Specifically, we examine possible changes in two dimensions of travel behavior: (i) mode choice; and (ii) number of trips. Using recent data (N=2,705) from the Public Attitudes Survey of Transport Scotland, statistical modeling was conducted in order to identify which population segments are more likely to change their travel mode and reduce the number of trips they make in response to the ongoing crisis. To account for possible, systematic unobserved effects that may underpin these two behavioral dimensions, the Bivariate Ordered Probit (BOP) framework was used for model estimation, thus enabling the joint modeling of two dependent variables. The survey data showed that 26.1% of the respondents changed their typical mode of travel to save money, and almost 38% of the respondents exhibit a propensity to reduce their number of trips due to rising travel costs. The results of the BOP model showed that low-income households and ethnic minorities are among the most severely affected socio-demographic groups, as they are more likely to have changed their travel mode and reduce their amount of travel. Women, people with disabilities, millennials, and residents in rural areas of Scotland also show evidence of behavioral change in light of the cost-of-living crisis. The findings of this study highlight major inequalities in transport accessibility, which will continue to deepen and result in more transport poverty, as the cost-of-living crisis unfolds, and appropriate remedial measures are not taken by legislative and governmental Authorities.

Keywords: Bivariate Ordered, cost-of-living crisis; mode of travel; number of trips; transport equity; travel behavior

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INTRODUCTION

The COVID-19 pandemic had a tremendous impact on the travel sector, not only due to the emergency measures that were in place across the globe, but also due to the significant changes in travel habits, which were accelerated by the elevated risk perception of travelers (Downey et al., 2022). In early 2022, the travel sector started to exhibit signs of recovery, mainly illustrated through a surge in travel demand for both short- and long-haul trips. However, the recovery trajectory was disrupted by another global challenge, which may have a substantial and long-standing impact on travel choices. This is the surge in global inflation, which can be partially attributed to the post-pandemic economic recovery and partially to the major Russia-Ukraine conflict.

In 2022, inflation has seen its highest levels over the last few decades, with the United Kingdom not being an exemption of this global phenomenon. Interestingly, the annual inflation rate surpassed 11% in October 2022, which constitutes a 41-year record high in the UK (*2*). The UK economy slipped into recession in the second half of the year 2022 and may likely remain in recession in 2023, as forecasted by the Bank of England. As expected, the sharp increase of inflation has had a major impact on living cost, primarily causing a significant drop in the purchasing power and disposable income of households – an effect referred to as "cost-of-living crisis" (*3*). Among the developed economies, the UK has been one of the most heavily affected by global inflation, with the crisis affecting all categories of businesses (small, medium, and large scale), rural and urban communities, and low-income households. In parallel with the global inflation, UK also faces additional economic challenges induced by the Brexit, which has particularly affected food price inflation and costing (*4*).

The rising inflation leads households to spend more on their basic needs, such as energy, housing, and food, and consequently, to cut back spending on other essential and non-essential services, including travel. Interestingly, one in three UK residents self-reported their willingness to reduce their travel-related expenditure due to cost-of-living concerns (*5*, *6*). The potential impact of the cost-of-living on travel plans is even more pronounced on British millennials, aged between 25 and 34 years old (*7*). Apart from the general increase of the living cost, the increase of transport costs is specifically cited as a significant barrier to Britons' travel choices (*8*). Focusing on commute travel, a recent study also showed that almost a third of Britons expect their commute to become soon unaffordable if the fuel prices continue to rise (*9*). Focusing on Scotland, four out of five people are highly concerned about the living cost crisis, whereas more than one in five struggle to afford their essential travel (*10*).

Previous evidence from across the globe suggests that in an era of economic crisis, the travel behavior is a lifestyle aspect heavily affected by the reduction of the household disposable income and associated purchasing power. This impact typically emerged through decreases in trip frequency (e.g., for shopping or leisure), changes in travel modes (e.g., shifts from car use to public transport or active travel) or even shifts in the perceived importance and utility of public transport (*11*). Interestingly, the 2008 financial crisis in Iceland led people to make fewer trips, with such reductions being more evident in non-urban population segments; however, low-income households were found to positively re-assess the utility of public transport (*11*). As a result of the long-standing Greek financial crisis, low-income households were recognized as more vulnerable to increasing travel costs, with many of these shifting from car to public transport (*12*). Following the 2008 financial crisis, Portuguese students experienced major shifts in their commuting patterns, primarily by using public transport more intensively (*13*). Overall, a common thread identified in the literature is associated with the equity dimension of the relationship between the soaring cost-of-living and travel choices. Low-income households, younger individuals, unemployed individuals, and generally underprivileged groups of the population are more likely to suffer the travel-related consequences and in turn, experience a higher risk of transport deprivation and social exclusion (*14*) due to limited, affordable travel options. The risi

Despite its utmost importance for economic and social prosperity, the effect of the ongoing living cost crisis on travel behavior has not been in-depth investigated to date. To address this gap, the main goal of this study is to assess whether and to what extent the rapidly rising living cost in Scotland, UK has affected travel choices of people, primarily in terms of trip frequency and mode choice. Furthermore, we seek to identify the determinants of the relationship between the cost-of-living crisis and travel behavior.

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The outcomes of the study will shed light on possible socio-demographic inequalities as to the access to transport provision, which are exacerbated by the crisis, and on emerging sources of transport poverty.

DATA

To identify the impact of the cost-of-living crisis on travel behaviour, data was drawn from the Public Attitudes Survey (PAS) of Transport Scotland (*15*). This survey has been carried out since May 2020, initially to track the impact of COVID-19 on travel and transport in Scotland. Following the ease of the COVID-19 restrictions, this survey temporarily ceased (in November 2021), but Transport Scotland restarted it in 2022 in order to gain insights into key aspects of travel behavior in the aftermath of COVID-19 and track public attitudes towards emerging, transport-related topics and issues. So far, 24 waves of the surveys have been carried out. PAS is a telephone-based survey, which is administered using both landline and mobile phone numbers, in line with the provisions of the General Data Protection Regulation (GDPR) and the Market Research Society (MRS) Code of Conduct. Through different waves of the PAS conducted at generally regular time intervals, different random samples of adults (over 16 years old, according to the Scottish legislation) across the entire Scotland were obtained. The sample is deemed overall representative of key strata of the Scottish population, as quotas are enforced with regards to the age, gender, region of Scotland and socio-economic classification (social grade) of the respondent's household.

In general questions change from wave to wave, so that information relevant to items high on the political agenda may be made available to decision-makers. Apart from questions relating to travel behavior and transport-related attitudes of Scottish residents, the survey elicits information about the socio-economic and demographic background of the respondent (e.g., age, gender, ethnic background, household income, social grade). The questions about travel behavior are mainly associated with mode choice currently and before the pandemic as well as travel frequency in light of the cost-of-living crisis. In addition, attitudes on the affordability of travel costs and its impact on travel choices are also obtained. In this study, data from the three most recent survey waves (22, 23 and 24) were compiled leading to a total sample of 3,021 responses. The timeline of the different survey waves along with information about inflation during the data collection periods are provided in the following Table:

TABLE 1 Data collection periods and evolution of inflation over time in the UK

Survey wave	N	Data collection period	UK Consumer Prices Index (CPI)*	Inflation rate for transport services*
Wave 22	878	16 May – 1 June 2022	9.10%	13.80%
Wave 23	913	13 July-26 July 2022	10.10%	14.80%
Wave 24	913	21 September – 9 October 2022	11.10%	8.90%

*Source: Office for National Statistics (ONS)

Due to missing information, the final sample used for statistical analysis consists of 2,705 responses. 48% of the sample (N=2,705) were identified as males, where 51% of the sample were identified as females. Both percentages are in accordance with the gender distribution of the Scottish population (*16*). 76.4% of the sample consisted of respondents aged between 16 and 64 years, whereas 20.6% of the respondents were 65 years or older – these statistics also closely resemble the age distribution of the Scottish population (*16*).

Two verbatim questions-statements, which provide insights into the self-reported implications of the cost-of-living crisis on travel choices of Scottish residents, serve as the key dependent variables of the statistical analysis in this study. These are:

(i) I have changed the mode of transport I use to save money.
(ii) I am no longer able to make as many journeys as I did due to travel costs.

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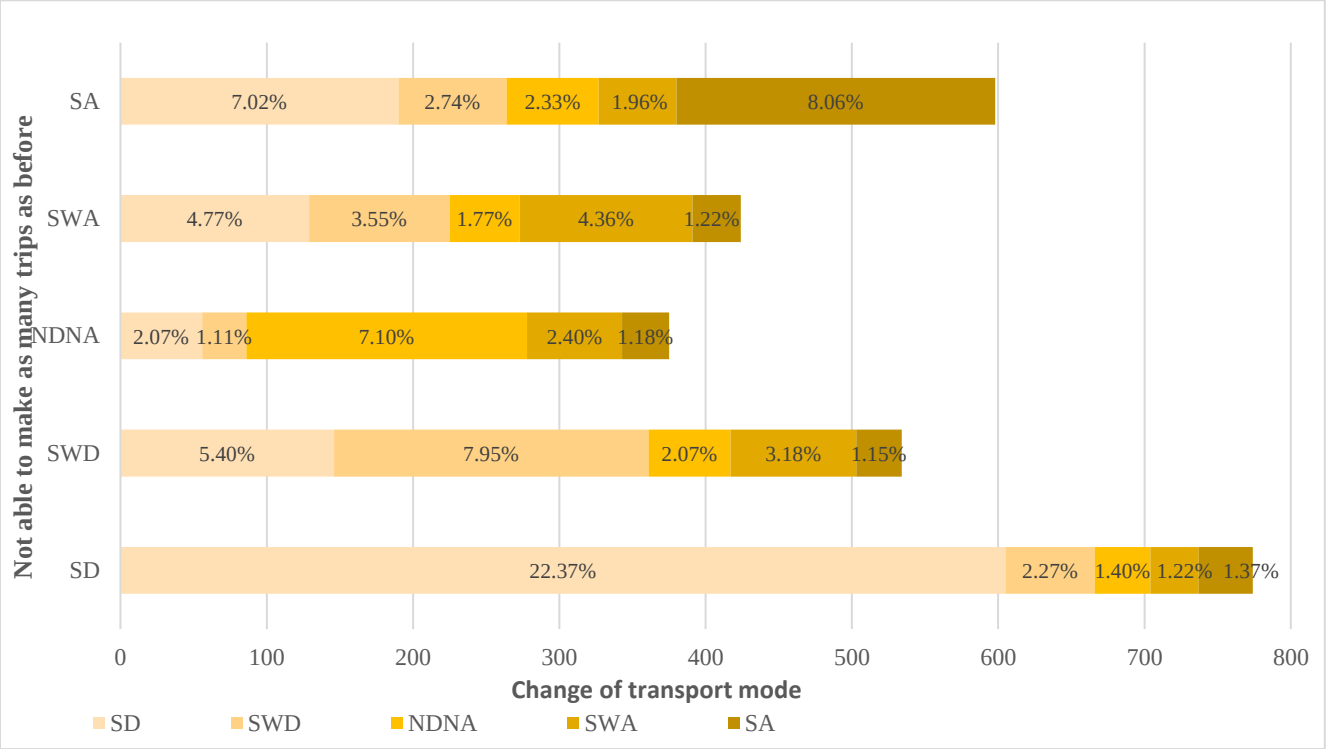
For both questions, the respondents were asked to rate their level of agreement on a 5-point Likert scale (ranging from 1: Strongly disagree to 5: Strongly agree). Figure 1 shows the joint distribution of the responses to the two questions:



SD: Strongly Disagree; SWD: Somewhat Disagree; NDNA: Neither Disagree Nor Agree; SWA: Somewhat Agree; SA: Strongly Agree

Figure 1 Distribution of responses (N=2,705) to key survey questions

As it can be inferred from Figure 1, 26.10% of the respondents strongly or somewhat agreed with the statement about the change of mode of transport, whereas almost 38% of the respondents provided evidence (through strong or slight agreement) on their inability to continue making the same number of trips as before, due to the increased travel costs. More than 15% of the respondents have reported both a change of transport mode and a tendency to make fewer journeys. This is the sub-sample whose travel habits have been disrupted the most by the cost-of-living crisis. Such disruption may be due to (a combination of) different reasons: on the one side, it may be due to fact that the mandatory expenses of the people in this group have increased the most due to inflation, leaving them with less money to spend on travel. On the other side, the reduction may indicate a greater flexibility of travel choices, possibly linked to a large prevalence of non-essential travel before the crisis. In contrast, almost four in ten respondents (37.92%) do not seem to have altered their travel behavior, as neither their mode choice nor their ability to maintain the same trip patterns have changed. Overall, among those making behavioral changes to address the implications of crisis, the reduction of trips is found to be more prevalent compared to change of transport mode. This disparity between these two behavioral aspects may not be a matter of choice, but – similar to what discussed above – the result of the availability of viable transport alternatives individuals have access to. The statistical modeling of both aspects will shed light on the factors determining the likelihood of each specific behavioral change.

METHODOLOGY

Due to their Likert-scale formulation, the dependent variables (i.e., the key survey questions) are ordered in nature, so to account for the inherent hierarchy of their outcomes, the ordered probit framework was

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employed for statistical modeling. The ordered probit model is formulated through a latent variable, y^*, which, for each respondent i, is defined as (*17*):

$$y_i^* = \boldsymbol{\beta}\mathbf{X}_i + \varepsilon_i \tag{1}$$

Where, $\mathbf{X}$ represents a vector of independent variables affecting the dependent variable, $\boldsymbol{\beta}$ is a vector of coefficients (parameters) associated with $\mathbf{X}$, and ε denotes an error term considered to vary according to the standard normal distribution (mean zero and variance one). The latent variable defines the probability of the outcome k for the individual according to the equation:

$$P_i(k) = \Phi(\mu_k - \boldsymbol{\beta}\boldsymbol{X}_i) - \Phi(\mu_{k-1} - \boldsymbol{\beta}\boldsymbol{X}_i), k\epsilon[0, Z-1] \tag{2}$$

Where Φ is the cumulative function of the standard normal distribution, μ_k are estimable parameters, and Z the number of possible outcomes. Following the formulation of (*18*), it is assumed that the μ_0 (i.e., the first threshold) takes a zero value, therefore k-2 thresholds constitute estimable parameters of the model, where k is the total number of outcomes of the dependent variable. $\boldsymbol{\beta}$ and μ_k are estimated so that the maximise the combined likelihood of observing all the measured outcomes.

Both dependent variables capture the reported impact of the rising living and travel costs on different, but closely related aspects of travel behavior (i.e., Mode Choice - MC and Number of Journeys - JN). Due to their conceptual proximity, common unobserved characteristics may determine responses about these two behaviors. Such unobserved effects could cause correlation of the error terms of the two latent functions. The conventional modeling approach, i.e., the estimation of separate, univariate ordered probit models for each dependent variable cannot address the issue of the cross-equation error term correlation (*18*), potentially leading to biased parameter estimates. To overcome this problem, the Bivariate Probit (BOP) framework is employed, which allows the joint modeling of two dependent variables, accounting, at the same time, for the possible correlation of the error terms. The model is formulated as:

$$\begin{aligned} y_{i,MC} *&= \boldsymbol{\beta}_{i,MC}\boldsymbol{X}_{i,MC} + \varepsilon_{i,MC} \\ y_{i,JN} *&= \boldsymbol{\beta}_{i,JN}\mathbf{X}_{i,JN} + \varepsilon_{i,JN} \end{aligned} \tag{3}$$

Where, $y_{i,MC}$* and $y_{i,JN}$* are the latent variables corresponding to the two jointly-modeled dependent variables (i.e., change of transport mode & tendency to make fewer journeys) and all other terms are as previously denoted. According to the Washington et al. (*18*), the error terms of the two dependent variables follow the bivariate standard normal distribution. The cross-equation correlation of the error terms is an estimable parameter of the model, defined as Tsavdari et al. (*19*):

$$\begin{pmatrix} \varepsilon_{i,1} \\ \varepsilon_{i,2} \end{pmatrix} \sim \mathcal{N}\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right] \tag{4}$$

Where, ρ represents the cross-equation error term correlation, and $\mathcal{N}$ is the bivariate normal distribution. The estimation of the bivariate ordered probit model was achieved using a full information maximum likelihood estimation approach (*19*).

To measure the impact of each independent variable on the likelihood of each outcome of the dependent variables, marginal effects were also calculated. Given that the estimated model consists only of indicator variables (see the "Results and Discussion" section for further details), the marginal effects quantify the change in the likelihood of the dependent variable due to a change in the value of an independent variable from zero to one. In the context of the ordered probit framework, marginal effects also illustrate the actual effect of the independent variables on the intermediate outcomes of the dependent variables, which cannot be readily provided by the parameter estimates of the model (*20*).

RESULTS AND DISCUSSION

The estimation results of the BOP model, along with key descriptive statistics of its independent variables are presented in Table 2, whereas the marginal effects of the model's independent variables are presented

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in Table 3. In general, all possible factors (and several interactions) that could potentially affect the dependent variables were trialed, with the presented model providing the best statistical fit in terms of log-likelihood at convergence and McFadden pseudo-R^2. The factors investigated throughout the modeling course are provided in the Appendix (Table A1) along with their descriptive statistics. All independent variables were detected to exert statistically significant effects, with the vast majority being significant at a greater than 95% level of confidence. A positive sign of the coefficient suggests an increase in the likelihood of the "Strongly Agree" outcome (i.e., denotes a stronger perceived effect of the living cost crisis on travel choices), whereas a negative sign implies an increase in the likelihood of the "Strongly Disagree" outcome. Likelihood Ratio Tests were conducted to confirm not only the significance of the BOP model compared to the null model (with zero variables), but also the superiority of the BOP against its univariate counterparts (i.e., a univariate ordered probit model for change of transport mode and a univariate ordered probit model for the tendency to make fewer journeys).

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TABLE 2 Estimation results of the BOP model

	Variable description	Percentage (%)	Change of travel mode		Making fewer trips	
			Coefficient	t-stat	Coefficient	t-stat
Constant			-0.202	-4.15	0.517	12.61
Socio-demographic characteristics	Gender (1 if female, 0 otherwise)	51.39%	-	-	0.180	4.84
	Millennial (1 if the respondent belongs to the millennial generation, 0 otherwise)	32.98%	0.084	1.99	-	-
	Baby boomer (1 if the respondent belongs to the baby boomer generation, 0 otherwise)	30.98%	-	-	-0.090	-2.19
	Low-income household (if the annual income of the respondent's household is less than £20,000, 0 otherwise)	7.87%	0.205	2.82	0.312	4.39
	AB Social grade (1 if the Household Reference Person works in a higher & intermediate managerial, administrative, or professional role, 0 otherwise)	26.65%	-	-	-0.195	-4.46
	Ethnic background (1 if ethnic minority group – i.e., any mixed, Asian, or Black background, 0 otherwise)	3.25%	0.314	2.61	0.240	1.91
	Health problem or disability (1 if day-to-day activities are limited a lot due to a health problem or disability, 0 otherwise)	11.72%	-	-	0.236	4.39
Current use and access to travel modes	Active travel (1 if walking/cycling is among the respondent's main modes of travel, 0 otherwise)	55.93%	0.318	7.98	-	-
	Public transport use (1 if bus/train/metro/ tram is among the respondent's main modes of travel, 0 otherwise)	38.23%	0.359	8.71	-	-
	Bicycle access (1 if the respondent's household has access to a bicycle, 0 otherwise)	51.61%	0.116	3.02	-	-
Spatial & temporal traits	Scotland main conurbations (1 if the respondent lives in Edinburgh and Southeast Scotland or in Glasgow City, 0 otherwise)	48.13%	0.063	1.63*	-	-
	Highlands and Islands (1 if the respondent lives in Highlands and Islands, 0 otherwise)	5.66%	-	-	0.186	2.39
	Survey wave (1 if 22, 0 otherwise)	32.38%	-0.133	-2.90	-0.101	-2.28
Thresholds & correlation	μ1		0.459	22.63	0.544	24.16
	μ2		0.877	32.32	0.894	32.61
	μ3		1.384	39.01	1.342	40.27
	Cross equation error term correlation (ρ)		0.518	31.65		
Goodness-of-fit	Number of observations		2,705			
	Log-likelihood at zero [LL(0)]		-8396.740			
	Log-likelihood at convergence [LL(**β**])		-7845.620			
	Akaike Information Criterion (AIC)		15741			

*****A Likelihood Ratio Test (LRT) was conducted for this variable and the results showed that the variable is statistically significant at a 0.90 level of confidence.

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TABLE 3 Marginal effects of variables included in the BOP model

	Variable description	Change of travel mode					Making fewer trips				
		SD	SWD	NAND	SWA	SA	SD	SWD	NAND	SWA	SA
Socio-demographic characteristics	Gender (1 if female, 0 otherwise)	-	-	-	-	-	-0.060	-0.011	0.003	0.015	0.052
	Millennial (1 if the respondent belongs to the millennial generation, 0 otherwise)	-0.049	0.000	0.008	0.015	0.026	-	-	-	-	-
	Baby boomer (1 if the respondent belongs to the baby boomer generation, 0 otherwise)	-	-	-	-	-	0.042	0.007	-0.003	-0.011	-0.036
	Low-income household (if the annual income of the respondent's household is less than £20,000, 0 otherwise)	-0.076	-0.003	0.012	0.023	0.043	-0.093	-0.024	0.000	0.021	0.096
	AB Social grade (1 if the Household Reference Person works in a higher & intermediate managerial, administrative, or professional role, 0 otherwise)	-	-	-	-	-	0.080	0.012	-0.006	-0.021	-0.065
	Ethnic background (1 if ethnic minority group (any mixed, Asian, or Black background), 0 otherwise	-0.118	-0.008	0.015	0.036	0.074	-0.074	-0.019	0.001	0.017	0.075
	Health problem or disability (1 if day-to-day activities are limited a lot due to a health problem or disability, 0 otherwise)	-	-	-	-	-	-0.091	-0.023	0.001	0.021	0.092
Use and access to travel modes	Active travel (1 if walking/cycling is among the respondent's main modes of travel, 0 otherwise)	-0.135	0.002	0.025	0.042	0.068	-	-	-	-	-
	Public transport use (1 if bus/train/metro/ tram is among the respondent's main modes of travel, 0 otherwise)	-0.132	-0.002	0.022	0.040	0.071	-	-	-	-	-
	Bicycle access (1 if the respondent's household has access to a bicycle, 0 otherwise)	-0.045	0.000	0.008	0.014	0.023	-	-	-	-	-
Spatial & temporal characteristics	Scotland main conurbations (1 if the respondent lives in Edinburgh and Southeast Scotland or in the Glasgow City, 0 otherwise)	-0.028	0.000	0.005	0.009	0.014	-	-	-	-	-
	Highlands and Islands (1 if the respondent lives in Highlands and Islands, 0 otherwise)	-	-	-	-	-	-0.049	-0.011	0.001	0.012	0.047
	Survey wave (1 if 22, 0 otherwise)	0.049	-0.001	-0.009	-0.015	-0.024	0.037	0.006	-0.002	-0.010	-0.032

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Low-income households are identified to be significantly affected by the cost-of-living crisis. Table 2 shows that individuals from low-income households (with annual income less than £20,000) are more likely (compared to households of higher income) to have changed their mode of travel to save money. The same population segment is also more likely to make fewer trips due to travel costs. This finding highlights the vulnerability of low-income households to rising living costs, as they may be compelled to cut back spending on services like transport in order to cover fundamental needs, such as food, energy and housing costs. Due to the inflation rates and their lower financial resilience, the proportion of their budget devoted to essential travel is already higher compared to financially more resilient socio-demographic groups (*10*). Therefore, the crisis has forced them to drastically modify their travel habits in an effort to cut back larger portions of their travel budget compared to other income strata. This trend is also identified by recent data from the Office of National Statistics in the UK (*5*) showing that the proportion of households that reduce the number of non-essential journeys made using their own vehicle is significantly higher for lower incomes. It should be noted that low-income households are typically inclined to use less their personal vehicles, with this behavior being evidenced even before the onset of the cost-of-living crisis (*21*). This population segment is expected to turn towards more affordable travel options, such as public transport and active travel means. The low-income variable has the strongest effect in the model component about the tendency towards making fewer trips. Specifically, Table 3 shows that if the respondent is a member of a low-income household, the likelihood of "strongly agree" or "somewhat agree" outcomes increases collectively by 11.7%.

As opposed to low-income households, the travel behavior of respondents belonging to the social grade A or B does not seem to be affected by the rising travel costs. The social grade is a socio-demographic classification used in the UK exclusively based on the type of the occupation of the Household Reference Person, with AB reflecting higher & intermediate managerial, administrative, and professional occupations. As shown in Table 2, these individuals are less likely to reduce their number of trips. This is an anticipated outcome as recent evidence (*22*) has shown that social grade AB has been the least affected by the cost-of-living crisis compared to all other grades (C1, C2, DE). Previous evidence has shown that this social grade may rely less on physical travel for work purposes, due to higher involvement in teleworking activities – this trend was strongly observed during the pandemic (*20*).

The impact of the cost-of-living crisis on travel behavior is also identified to vary across different ages. Specifically, the need of millennials (born between 1981 and 1996) to change their mode of travel in order to reduce their transport-related expenditure is clearly documented in the results of the BOP model. In contrast, baby boomers (born between 1946 and 1964) are less likely to reduce the number of trips they make. The rising living cost constitutes a top concern for millennials (*23*) and given their lower dependence on car-oriented lifestyles (*24*) compared to earlier generations, they may be more willing (and able) to switch to affordable travel modes (e.g., active travel, public transport, or shared mobility services), especially if the latter are considered as environmentally sustainable. According to recent evidence, baby boomers in the UK are less affected by the rising inflation compared to other generational strata (*25*), and as expected, they are more reluctant to make lifestyle changes and reduce their mobility patterns. These results reveal generational disparities in the impact of the cost-of-living crisis, which need to be carefully addressed in the future.

Table 2 shows that females are also more likely to reduce the number of trips compared to males, due to rising travel costs. According to a recent study of the Scottish Government, women are more exposed than men to the growing burden of the living cost, mainly due to their lower wages or savings, more caring responsibilities and their potential involvement in economic sectors most severely affected by the crisis (*10*). The propensity of female respondents towards fewer trips may be attributed either to the crisis-led increase of the cost of owing or driving a car, which makes the car use unaffordable for a major portion of women, or their lack of access to adequate public transport services, especially in remote or rural areas of Scotland (*26*).

Ethnic minorities are also detected as one of the most vulnerable groups, with respondents of any mixed, Asian, or Black background being more likely to have changed their mode of travel since the onset of the cost-of-living crisis. The same population segment is also more likely to reduce the overall number

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of trips made. Notably, the specific variable has the strongest impact on the model component about change of travel model, with Table 3 showing that for individuals of this group, the likelihood of the "strongly agree" outcome increases by 7.4%. These results re-iterate the well-documented vulnerability of minority ethnic households not only to the cost-of-living crisis (as recently documented by the Scottish Government, (*10*)), but also to both energy and transport poverty (*27*). Due to their relatively higher likelihood to face financial hardships in combination with soaring inflation, minority groups may need to spend higher proportions of their income for their essential needs, thus cutting back expenses for travel. In addition, some minority groups may be also living in areas of lower housing costs where the provision of transport services may not be adequate (*27*), above all for non-vehicle owners, thus limiting considerably their mobility options.

Another socio-demographic group that is disproportionately affected are those with a health problem or disability. The latter are also shown to be more inclined to reduce their amount of travel due to rising costs. These individuals typically face significant challenges in terms of transport accessibility, and as a result, it is estimated that they make 38% fewer trips than individuals without disabilities, even before the onset of the crisis (*28*). This accessibility gap is likely to be exacerbated by the living cost soar because households with members with a disability need to overspend on personal transport (33% according to the latest figures), mainly due to accessibility issues of public transport (*29*) and active travel infrastructure.

The results of the BOP models also suggest that those currently using public transport or active travel means are more likely to have recently changed their mode of travel in an effort to reduce their transport-related expenses. This result reflects the eagerness of individuals to identify and use more affordable travel options and it is also backed up by a recent survey across the entire UK (*6*) showing that 72% of respondents opted for cycling and walking for short trips they used to make by their personal car or public transport. Particularly, in Scotland, 64% of the respondents switched to public transport; that switch was deliberately made as a cost-saving measure. These findings show that the rising transport costs could potentially serve as an accelerator of behavioral change if these changes are stabilized by means of adequate transport policies. Policymakers should take into account these trends while designing policies and investments to ensure equitable provision of public transport services and active travel infrastructure. Similarly, members of households with access to a bicycle are also associated with a greater likelihood of mode choice shift, possibly towards cycling. For these individuals, opting for active travel is even more feasible, as there is direct access to the mode.

Focusing on location-specific characteristics, higher likelihood of mode choice change is also observed for the residents of the region of Edinburgh and Southeast Scotland and of Glasgow city, as shown in Table 2. These regions form a major part of Scotland's central belt, which has the highest population density and encompasses the two largest urban areas (Glasgow and Edinburgh) in the country. In these areas, due to potentially greater availability of transport services and infrastructure, the switch to more affordable travel modes (e.g., public transport, shared mobility services or active travel) as a way to save money may be more feasible, especially compared to more remote areas. In fact, the BOP results show that residents of the Highlands and Islands are more prone to a reduction of their trips, because of the soaring travel costs. Highlands and Islands is a representative example of a rural and remote area of Scotland, where the fuel poverty levels are among the highest in the country (*30*). Rural and island households in Scotland tend to spend more for transport services (compared to their urban counterparts) due to higher fuel prices, the longer distances they need to cover as well as the limited provision of public transport (*10*). The latter constitutes a major issue for rural residents, as the available alternatives to car use are extremely limited, thus prompting them to respond to the surge in fuel/energy costs by making less trips, and effectively constraining their mobility patterns.

Lastly, the fluid economic environment, proxied by the timing of the survey waves, was found to be associated with respondents' behavioral changes. Specifically, those that were interviewed during the survey wave 22 (conducted in May 2022) were less likely to report any change in terms of both mode choice and number of trips compared to the other two survey waves (23 and 24). As shown in Table 1, the overall inflation rate (expressed through the Consumer Prices Index) during the wave 22 period was high in absolute terms, but relatively lower than the inflation rate in the next two survey waves. It is likely that

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during the later survey waves, the respondents had already experienced the consequences of the cost-of-living crisis, which forced them to modify their travel choices for the sake of budget savings. During the wave 22, respondents were naturally concerned about the cost-of-living crisis, but these concerns had not been translated into lifestyle changes. It should be noted that some remaining COVID-19 restrictions were lifted in Scotland just a few weeks before the period of wave 22, so it is likely that the respondents were still reluctant to proceed to changes that would further limit their mobility patterns.

Table 2 also shows that the cross-equation error term correlation is statistically significant (at a 99% level of confidence). That means the error terms corresponding to the two dependent variables of the BOP model are indeed correlated, thus providing statistical evidence on the suitability of the bivariate modeling technique. The significance of the correlation proves that the unobserved characteristics captured by the error terms indeed share systematic similarities in the way they affect both dependent variables. Such systematic similarities of the unobserved characteristics have been accounted for throughout the estimation of the bivariate probit model, and such, the parameter estimates of the independent variables are more robust (*31*). The positive sign of the correlation coefficient implies that the common unobserved characteristics affect both dependent variables in the same direction; in other words, unobserved characteristics that favor a change in the travel mode, they also favor an increase in the propensity of the reduction of the number of trips, and vice versa.

POLICY IMPLICATIONS AND CONCLUSIONS

In this study, we sought to understand how and to what extent the ongoing cost-of-living crisis affects the travel behavior of Scottish residents by focusing on two fundamental dimensions: mode choice and number of trips. To that end, data from several waves of the Public Attitudes Survey of Transport Scotland were used and statistically analyzed. To account for the potential impact of shared unobserved effects between these inter-related behavioral dimensions, a bivariate ordered probit model was estimated, which allowed for the joint modeling of both behavioral aspect within an integrated estimation framework. The descriptive analysis of the data showed that more than one in four respondents reported a change in their main travel mode, whereas almost four in ten mentioned they are no longer able to make as many trips as before the onset of the cost-of-living crisis.

The BOP results suggested that low-income households and ethnic minority groups are among the most severely affected population segments of the crisis, as they are both associated with a higher propensity towards a change of their travel mode and towards making fewer trips in response to the rising costs. Females and those with a health problem or disability are more likely to reduce their amount of traveling due to the crisis, whereas baby boomers and respondents engaged in higher & intermediate managerial, administrative, and professional occupations are identified as more resilient socio-demographic groups. Millennials and residents of the major urban areas of Scotland show a higher tendency to change their mode of travel, whereas for the residents of Highlands and Islands, the rising travel cost may prompt them to reduce their amount of trips.

The findings of this study shed light on major inequalities as to the impact of the cost-of-living crisis on travel behaviour, and general transport accessibility. These inequalities are not new, but they are further deepening in light of the crisis. In a nutshell, low-income households, ethnic minorities, disabled people, younger individuals and females, as well as rural households are disproportionally affected by the crisis (compared to other socio-demographic groups). If remedial measures are not taken soon by the central government and local Authorities, these inequalities will further deepen, putting the aforementioned groups at greater risk of transport poverty and, in turn, social exclusion. These measures should focus on ensuring affordable and accessible travel options for those at higher risk through concessionary ticket schemes and fare caps for public transport services. During times of economic crisis, public transport should be leading the way towards more affordable, but sustainable travel, especially for the under-privileged groups; as such, higher public investment should be directed towards the upgrade of the public transport provision and improvement of their quality of service (*32*), especially in rural areas, where the services are quite sparse. In these areas, where the lack of infrastructure may hamper a rapid recovery of public transport, which has been strongly hit by the COVID-19 pandemic (*1*), the use of formal and informal shared mobility services

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could be further promoted through digital platforms, Mobility-as-a-Service (MaaS) schemes and user incentives. Improvement of provision for public transport and shared mobility services should be also prioritized in deprived suburbs of major urban areas, where the most disadvantaged groups face the most pronounced mobility barriers. As suggested by the findings of the study, active travel may serve as an affordable and sustainable alternative to motorized modes (either by users' deliberate choice or not); hence, it is suggested that investments on active travel infrastructure (e.g., cycle and pedestrian paths, segregated routes) should be intensified. Lastly, community transport programs that can be tailored to the needs of those with disabilities or carers or those located in remote areas can take the pressure off the budget of socio-demographic groups that a priori need to pay extra costs to accommodate their special travel requirements.

Overall, the prioritization of public transport and active travel may not only help narrow the inequalities on transport accessibility induced by the cost-of-living crisis, but it may also facilitate the achievement of the net-zero target in the transport sector, which is clearly documented in Scotland's National Transport Strategy (*33*). In that way, the threat posed by the economic crisis can be reversed into an opportunity for impactful climate action and equitable mobility for all.

The data available for our analysis clearly suggest that cost of living crisis has impacted travel behavior and that such impacts are not distributed uniformly across the population, which is alarming enough to require careful attention from policy makers and all central/local Authorities. However, the dataset we used does not provide information on the extent of the change, on the type of trips impacted, and on the direction of the modal shift. Further research is required to clarify these aspects and so fine-tune and prioritize interventions.

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AUTHOR CONTRIBUTIONS

The authors confirm contribution to the paper as follows: study conception and design: A.F. and G.F.; data collation: A.O.; analysis and interpretation of results: G.F., A.F. and A.O.; draft manuscript preparation: G.F., A.O., and A.F. All authors reviewed the results and approved the final version of the manuscript.

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APPENDIX

TABLE A1 A synopsis of the variables of the entire sample explored in the statistical analysis (N=3,021)

Variable	Description	Outcomes	Percentage (%)
Changed mode of transport to save money	I have changed the mode of transport I use to save money	Strongly disagree	41.89%
		Somewhat disagree	17.74%
		Neither agree nor disagree	14.56%
		Strongly agree	12.97%
		Somewhat agree	12.84%
Changed journey frequency due to travel cost	No longer able to make as many journeys as I did due to travel costs	Strongly disagree	29.06%
		Somewhat disagree	21.98%
		Neither agree nor disagree	19.54%
		Strongly agree	15.63%
		Somewhat agree	13.78%
Gender	Gender	Female	51.29%
		Male	47.95%
		Prefer not to say	0.45%
		Prefer to self-describe	0.17%
		Non-Binary	0.14%
Public transport use	Public transport mode use, which include Underground, metro, light rail, tram, train & Bus, minibus or coach	Use public transport	61.70%
		Don’t use public transport	38.30%
Active travel	Active travel users which include bicycle, E-Bike & walking.	Use active travel	56.04%
		Don’t use active travel	43.96%
Age	Age group	Generation Z	5.90%
		Millennials	31.98%
		Generation X	19.63%
		Baby Boomers	31.35%

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		Other Age Groups	11.09%
Region location	Region of Scotland you live in	Argyll & Bute	1.72%
		Ayrshire & Arran	7.08%
		Edinburgh and South East Scotland	15.66%
		Forth Valley	5.56%
		Glasgow City	32.90%
		Highlands and Islands	5.73%
		North East Scotland	10.72%
		Scottish Borders	2.18%
		South West Scotland	2.88%
		Tay Cities Region	14.63%
		Prefer not to say	0.93%
Access to car/van	Do you/your household have access to a car or van for private use?	Yes	84.91%
		No	14.99%
		Don't Know	0.10%
Access to bike	Do you/your household have access to a bike suitable for adults?	Yes and roadworthy	46.11%
		Yes but not roadworthy	4.53%
		No	48.89%
		Don't know	0.46%
Social Group	Social Group	AB	26.38%
		C1	27.08%
		C2	21.71%
		DE	19.23%
		Refused	5.59%

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Health/Disability	Are your day-to-day activities limited because of a health problem or disability which has lasted, or is expected to last, at least 12 months?	Yes, limited a lot	11.98%
		Yes, limited a little	8.87%
		No	76.73%
		Prefer not to say	2.42%
Income	Household income categories before tax and other deductions	Low Income	16.69%
		Medium Income	41.29%
		High Income	42.02%
National Identity	Category which best describes your national identity	Scottish	65.97%
		English	2.42%
		Welsh	0.23%
		Irish	0.79%
		British	25.62%
		Other (please specify)	2.71%
		Prefer Not to say	2.25%

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